

Institutional Ethics & Participant Consent Statement

Study: Explainable Machine Learning for Student Burnout Classification and Risk Stratification: A Mixed-Methods Study with Qualitative Triangulation

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1. Ethical Standards and Study Framework

This study was conducted in accordance with the ethical principles outlined in the **Declaration of Helsinki** for research involving human participants.

- **Study Design:** Minimal-risk observational cross-sectional survey (\$N = 601\$) and voluntary semi-structured qualitative interviews (\$N = 20\$).
- **Data Collection Period:** Quantitative survey — completed **June 2026**; Qualitative interviews — conducted **June–July 2026**.
- **Target Population:** Enrolled university students in Bangladesh across 11 institutions (8 private universities, 1 public university, 2 National University-affiliated colleges). All participants were verified adult enrolled university students (18+ years).
- **Ethics Exemption Status:** The study was determined to be **exempt from formal ethics committee review** under the criteria applicable to low-risk, non-interventional, observational educational research. Ethics oversight is attributed to the **Department of Computer Science and Engineering, Presidency University, Dhaka-1212, Bangladesh**.

Ethics exemption documentation is available from the corresponding author (rifatmiah1992003@gmail.com) upon reasonable request.

2. Participant Informed Consent

Quantitative Survey Strand (\$N = 601\$)

- Prior to completing the survey instrument via Google Forms, all participants were presented with an electronic Informed Consent Form detailing:
 1. The educational and empirical research goals of the study.
 2. The strictly voluntary nature of participation, with freedom to exit at any time without academic consequence.
 3. De-identification measures ensuring full confidentiality.
- Electronic consent was mandatory before accessing survey questions.

Qualitative Interview Strand (\$N = 20\$)

- Qualitative interviews were conducted only after obtaining explicit voluntary informed consent.
 - Participants were assigned standardized pseudonyms (e.g., *Participant 1 [P1]*, *Participant 20 [P20]*, *Private University A*, *National University System*) to protect participant identities.
 - No personally identifying information was recorded or published.
 - All participants were informed of their right to withdraw at any time with no academic penalty.
 - Contact details for university counselling support were provided to all interview participants.
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3. Data Protection and Anonymization

- All primary survey responses were fully de-identified (*Quantitative_Survey_Data.xlsx*).
 - Qualitative transcripts were anonymized and stored on password-protected, encrypted local storage accessible solely to the primary research team.
 - Publicly deposited repository artifacts retain full anonymity in compliance with open science and data privacy standards.
 - **Data Availability:** Anonymised dataset, COREQ checklist (Supplementary S2), TRIPOD checklist (Supplementary S3), and interview guide (Supplementary S1) are available at: <https://github.com/rifatmiah92/student-burnout-ml-mixed-methods>
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4. Institutional Acknowledgement

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For any correspondence regarding ethics documentation, contact:

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